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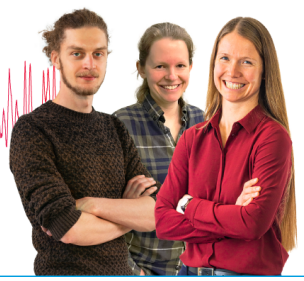
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Dissociation energies of ^{7}LiD*

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The $B\,^1\Pi - X\,^1\Sigma^+$ absorption spectrum of ^{7}LiD has been reexamined. Evidence for an additional vibrational level ($v'=3$) in the upper state has been found. Because this vibrational level is extremely close to the dissociation limit, only $J'=1$ and $J'=2$ levels occur. This additional information provides an improved dissociation limit of 34671.5 ± 0.4 cm^{-1}, 1.5 cm^{-1} lower and 20 times more precise than previous results. From this result, dissociation energies of 19767.7 ± 0.4 cm^{-1} for the X state, 8588.4 ± 0.9 cm^{-1} for the A state, and 201.8 ± 0.4 cm^{-1} for the B state of ^{7}LiD are derived.

INTRODUCTION

Even though lithium hydride has been called "the workbench of the theoretical chemist" by Bender and Davidson,[1] remarkably little electronic spectroscopic work on the isotopic lithium hydrides has been carried out since the 1930's.[2-4]

Recently, the $B^1\Pi - X^1\Sigma^+$ absorption spectrum of ^{7}LiH has been reanalyzed to obtain a more accurate potential curve and dissociation limit than had previously been available.[5] In the course of this analysis, it became obvious that an additional vibrational level ($v'=3$) not previously reported[3] must exist for ^{7}LiD. Moreover, observation of this additional level, predicted to be quite weakly bound, would be expected to provide a more accurate dissociation limit for ^{7}LiD given an analysis such as reported for ^{7}LiH in Ref. 5. These predictions have in fact been borne out by experimental observations, and the new observations have in turn been analyzed to obtain improved dissociation energies for ^{7}LiD.

PREDICTIONS

The existence of a $v'=3$ level can be directly predicted from the mass-reduced analysis[6] briefly described in Ref. 5. However, since a fairly accurate potential curve (G of Ref. 5) is available for the $B\,^1\Pi$ state, one can directly calculate (See Table I) the expected eigenvalues for $J'=1$ and 2 from the radial Schrödinger equation.[7] The $J'=3$ level is predicted not to exist as its "energy" (estimated by extrapolation) is well above the centrifugal barrier; i.e., rotational predissociation occurs.

EXPERIMENTAL OBSERVATIONS

The B–X absorption spectrum[3] has been reexamined and 4 lines of the 3–1 band clearly identified (Table II) with estimated uncertainties of ±0.04 cm^{-1} as in previous work.[3] As previously noted[3] for other bands in this system, the Q branch is most intense, the R branch weaker, and the P branch still weaker. In particular, the $P(2)$ and $P(3)$ lines were not clearly evident above the noise, as in the previously reported[3] 2–0 and 2–1 bands of ^{7}LiH and the 2–0 band of ^{7}LiD. This is in accord with the Hönl–London factors for these transitions,[8] which predict the following intensity ratios:

$$P(2) : Q(1) : R(0) = 1 : 3 : 2 \quad ,$$

$$P(3) : Q(2) : R(1) = 2 : 5 : 3 \quad .$$

The corresponding 3–0 band is weaker than the 3–1 band and has not yet been conclusively identified.

The term values corresponding to the assignments are given in Table I. The following molecular constants for the $v'=3$ level of the B state are obtained:

$$\nu_0(3-1) = 33640.6\ \text{cm}^{-1} \quad ,$$

$$B_3 = 0.32\ \text{cm}^{-1} \quad ,$$

although since the rotational constant is derived from a single spacing, and since D_3 is probably appreciable,[9] its value should not be taken very seriously. Utilizing these data in the determination of rotational and vibrational constants, we have

$$B_v(Q\ \text{lines}) = 1.904 - 0.425(v+\tfrac{1}{2}) - 0.005(v+\tfrac{1}{2})^2 - 0.001(v+\tfrac{1}{2})^3$$

$$G_v = 178.70(v+\tfrac{1}{2}) - 29.13(v+\tfrac{1}{2})^2 - 3.51(v+\tfrac{1}{2})^3$$

that may be compared with the constants previously given.[3]

DISCUSSION

The energy level results shown in Table I are in good agreement when the uncertainty ($\sim\pm2$cm^{-1}) in the poten-

TABLE I. Predicted B state of ^{7}LiD eigenvalues $E(v'=3, J')$ in cm^{-1} (measured with respect to the B state zero point level $E(0,1)$ are compared with the corresponding observed term values (see Table II).

J'	$E(v'=3, J')$ Predicted	Observed[a]
1	199.5	200.60
2	200.6	201.87

[a]The corresponding energies with respect to $v''=0$, $J''=0$ in the X state are 34670.19 and 34671.46 cm^{-1}, respectively.

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TABLE II. The 3–1 band of the $B-X$ system of ^{7}LiD.

J	$P(J)$	$Q(J)$	$R(J)$
0			33641.23
1		33633.04	33634.37
2	•••	33617.94	
3	•••		

tial energy curve used in the eigenvalue calculations is taken into account.

The dissociation limit may be readily predicted on the same basis as in Ref. 5. The $J'=2$ and 3 centrifugal barriers, respectively, are 0.27 ± 0.02 and 0.75 ± 0.04 cm^{-1} high and occur at 24.1 and 20.8a_0, clearly even farther into the long range region than the relevant barriers[5] for ^{7}LiH. Since no broadening of the $J'=2$ level is seen, this reference level must be at most 0.2 cm^{-1} above the dissociation limit. Since the line-like continuum structure corresponding to $J'=3$ (like that in Fig. 4 of Ref. 5) is not seen experimentally, one may say that the $J'=2$ level should probably not be bound by more than 0.3 cm^{-1} (corresponding to the potential G of Ref. 5) and certainly should not be bound by more than 0.4 cm^{-1}. Combining this result with the experimental energy of the $J'=2$ level, 34671.46 ± 0.04 cm^{-1}, obtains a dissociation limit of 34671.5 ± 0.4 cm^{-1}, which may be compared with the previous[3] estimate 34673 ± 8 cm^{-1}. This dissociation limit corresponds to the following D_0 values in cm^{-1}: 19767.7 ± 0.4 for the $X\,^1\Sigma^+$ ground state, 8588.4 ± 0.9 for the $A\,^1\Sigma^+$ state,[10] and 201.8 ± 0.4 for the $B\,^1\Pi$ state.[11] Finally, it should be noted that these D_0 values may be combined with estimates of the zero-point energies (e.g., from Ref. 3) to estimate D_e. However, a theoretical reanalysis of the regions near R_e in the X, A, and B states is now in progress, which should yield better estimates of the zero-point energies and their uncertainties for both ^{7}LiH and ^{7}LiD. This should make possible an accurate examination of the degree of breakdown of the Born–Oppenheimer approximation for comparison with theoretical estimates.[12]

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‡Phillips Petroleum Research Fellow (1972–1973).

[1]C. F. Bender and E. R. Davidson, J. Chem. Phys. 49, 4222 (1968).

[2]The only published electronic spectroscopic work we are aware of are Refs. 3 and 4 below, although new work is in progress at both The Institute of Optics and The University of Iowa.

[3]R. Velasco, Can. J. Phys. 35, 1204 (1957).

[4]I. Fernandez-Florez and R. Velasco, Opt. Pura Apl. 2, 123 (1969).

[5]K. R. Way and W. C. Stwalley, J. Chem. Phys. 59, 5298 (1973).

[6]A more complete analysis by one of us (WCS), using the isotopic mercury hydrides as an example, is in progress. Preliminary results were presented as Paper AA5, "The Utility of Mass-Reduced Quantum Numbers", at The 27th Symposium on Molecular Structure and Spectroscopy, Columbus, Ohio, June 1972.

[7]The authors wish to thank Prof. K. Sando for making available his eigenvalue program, descirbed in K. Sando, Mol. Phys. 23, 413 (1972).

[8]G. Herzberg, *Spectra of Diatomic Molecules* (Van Nostrand, Princeton, NJ, 1950), 2nd ed., p. 208.

[9]The eigenvalue calculations indicate $D_3\simeq0.006$ cm^{-1}.

[10]The $v=0$ level of the A state is not directly observed (only $v\geq1$), so an additional ±0.5 cm^{-1} is included.

[11]Note that D_0 for a Π state is measured from the zero-point level $E(v=0, J=1)$ to dissociation.

[12]P. R. Bunker, J. Mol. Spectrosc. 28, 422 (1968).

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